

Rapid MDRO Diagnostic Stewardship Public Artifact Suite

Author: Marious Akugri

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This public reference package is laboratory-scoped, non-prescriptive, and based on public guidance and synthetic examples. It is not medical advice, treatment guidance, institutional policy, or an approved clinical protocol.

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Implementation Map And Release Notes

Title: Rapid MDRO Diagnostic Stewardship Workflow and LIS Alert/Comment Reference for High-Complexity Clinical Laboratories

Author: Marious Akugri

Version: 0.4 reviewer-ready draft

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Purpose

This implementation map explains how the public artifact suite fits together. It is included so reviewers can evaluate the package as a practical laboratory-facing reference workflow rather than a set of disconnected draft documents.

The suite remains a public, non-confidential, laboratory-scoped reference package. It is not medical advice, treatment guidance, institutional policy, or an approved clinical protocol.

Version 0.4 is intended for outside technical review. The word "validation" in this package refers to reviewer dry-run assessment, workflow coherence review, and synthetic scenario testing. It does not mean clinical assay validation, institutional implementation, or validation of patient-care outcomes.

Package Map

Suite Component	Primary Function	Practical Use In A Laboratory Review
Public technical note	Defines scope, boundary, audience, and source basis.	Helps reviewers understand what the package is and what it is not.
Decision tree	Converts a rapid MDRO-related trigger into verification, routing, comment, follow-up, and documentation steps.	Gives staff and reviewers a single workflow view.
Reflex and confirmatory testing matrix	Maps common trigger types to result status, possible confirmatory review, communication action, and caveats.	Helps prevent preliminary, pending, final, and corrected results from being treated the same way.
LIS alert/comment template bank	Provides neutral status-aware comment examples.	Gives laboratories adaptable wording examples for local governance review.
TAT and escalation metric plan	Defines practical measures for notification, final-result follow-up, training, and feedback incorporation.	Helps convert the workflow into auditable quality indicators.
Staff training checklist	Provides training topics and synthetic case exercises.	Supports orientation, competency discussion, or mock review.
Result-status taxonomy	Defines common terms used across the suite.	Reduces ambiguity in communication and reviewer feedback.
Synthetic validation scenarios	Provides dry-run scenarios using non-patient examples.	Lets reviewers test whether the workflow handles realistic edge cases.
Reviewer feedback checklist	Gives reviewers a structured way to assess the package.	Supports targeted technical review before any public release or local adaptation.

Minimal Local Adaptation Path

A laboratory considering this package would need its own local review before use. A reasonable adaptation sequence may include:

1. Confirm the local scope: microbiology, molecular diagnostics, AST, C. auris screening, corrected-result workflows, or another approved subset.
2. Map local result statuses and communication categories to the result-status taxonomy.

3. Review the decision tree against existing laboratory director, medical director, quality, infection prevention, antimicrobial stewardship, and LIS governance requirements.
4. Select only the LIS comment templates that fit local policy and revise them locally before any use.
5. Test the workflow using synthetic or approved training examples before any live-system change.
6. Track a small set of metrics, such as notification documentation completeness, final-result follow-up completion, and staff training completion.
7. Review feedback and revise the local implementation under institutional governance.

Recommended Reviewer Path

For technical review, the most efficient path is:

1. Read this implementation map and the public technical note to understand the scope boundary.
2. Review the result-status taxonomy before reading the decision tree and matrix.
3. Use the synthetic scenarios to dry-run whether preliminary, final, corrected, pending-AST, discrepant, and *C. auris*-related findings are handled clearly.
4. Compare the generated or written LIS comments against the wording controls in the template bank.
5. Complete the reviewer feedback checklist or provide a short review memo identifying strengths, limits, and recommended revisions.

The reviewer does not need to endorse every sample phrase. The core review question is whether the package is coherent, laboratory-scoped, non-prescriptive, and useful as a public reference for local governance review.

Version History

Version	Status	Change Summary
0.1 reviewer draft	Preserved draft	Initial generated reviewer package.
0.2 reviewer draft	Preserved draft	Strengthened scope boundaries, source basis, and core workflow artifacts.
0.3 strengthened reviewer draft	Preserved draft	Adds this implementation map, result-status taxonomy, synthetic validation scenarios, and reviewer feedback checklist.
0.4 reviewer-ready draft	Current reviewer draft	Tightens the validation/adoption boundary, clarifies that review is technical rather than clinical validation, adds a clearer reviewer path, and prepares the package for named reviewer circulation.
1.0 public release	Planned	To be prepared only after technical review, revision, and final non-confidential safety review.

Evidence Boundary

This suite is designed to show a concrete, reviewable public artifact. It does not claim:

- clinical validation;
- institutional adoption;
- approval by any employer, hospital, public-health agency, or professional society;
- use with patient records;
- use with employer-confidential procedures;
- replacement of laboratory director or medical director approval; or
- authority to prescribe treatment, isolation, reporting, or transfer decisions.

Reviewer Orientation

Reviewers should evaluate whether the suite is technically coherent, practically useful, appropriately scoped, and safe to circulate as a public reference package. Reviewers should not treat it as a final SOP, a

treatment protocol, or an institutional policy document.

Suggested reviewer lanes for this version are: microbiology/AMR workflow review, public-health diagnostic communication review, and quality/regulatory documentation review. A reviewer only needs to address the lane they are qualified to assess.

Public Technical Note

Title: Rapid MDRO Diagnostic Stewardship Workflow and LIS Alert/Comment Reference for High-Complexity Clinical Laboratories

Author: Marios Akugri

Version: 0.4 reviewer-ready draft

Date: 2026-08-25

Summary

This technical note defines a public, non-confidential reference package for laboratory-facing rapid MDRO diagnostic stewardship. The package provides reusable workflow, communication, confirmation, escalation, and documentation examples for high-complexity laboratories handling MDRO-relevant results.

The package is not a treatment guideline, institutional policy, public-health reporting system, or replacement for medical, laboratory, infection prevention, antimicrobial stewardship, or regulatory oversight. It is a practical reference framework that laboratories may review and adapt under their own local protocols.

For this reviewer-ready version, "validate" means technical review, dry-run testing against synthetic scenarios, and revision based on qualified feedback. It does not mean clinical assay validation, proof of institutional adoption, or evidence that the package has been used with patient data.

Intended Audience

This reference package is intended for:

- medical laboratory scientists and clinical laboratory technologists;
- microbiology and molecular diagnostics personnel;
- laboratory managers and quality personnel;
- laboratory directors and technical supervisors;
- infection prevention collaborators;
- antimicrobial stewardship collaborators; and
- health-system personnel involved in diagnostic communication workflows.

The package is written from the laboratory perspective. It focuses on how laboratory result information can be verified, routed, documented, and followed up in a disciplined way.

Terms

AMR: antimicrobial resistance.

AST: antimicrobial susceptibility testing.

C. auris: *Candida auris*, an emerging multidrug-resistant fungal pathogen that can spread in healthcare settings.

LIS: laboratory information system.

MDRO: multidrug-resistant organism.

Rapid MDRO signal: a laboratory result or preliminary finding that suggests possible MDRO relevance, such as a rapid molecular panel result, organism-identification concern, resistance-marker detection, resistant AST phenotype, suspected or confirmed C. auris finding, or corrected/final result that changes the communication status.

Problem Statement

Rapid diagnostic testing can shorten the time between specimen collection, organism or resistance-marker detection, and communication to clinical teams. That speed is useful only if the result is handled with a clear

laboratory workflow.

In practice, rapid MDRO-related findings may involve several points of uncertainty:

- the result may be preliminary rather than final;
- culture and susceptibility testing may still be pending;
- organism identification may require confirmation;
- the result may require infection prevention or stewardship routing under local policy;
- a corrected or final result may require follow-up communication;
- staff may need consistent language for LIS comments and alerts; and
- communication documentation may vary across shifts, sections, or facilities.

This package addresses that workflow gap. It does not attempt to decide clinical treatment. Instead, it supports a consistent laboratory communication layer around rapid MDRO-related findings.

Purpose Of The Reference Package

The purpose of this package is to provide adaptable, non-confidential examples for:

- identifying rapid MDRO-related laboratory triggers;
- checking whether results are preliminary, final, corrected, or pending confirmation;
- routing laboratory communication to the appropriate institutional teams;
- documenting notification and follow-up actions;
- using neutral LIS comment language;
- connecting rapid results with reflex or confirmatory testing logic; and
- training laboratory staff on the distinction between laboratory communication and clinical decision-making.

Scope

This package is limited to laboratory diagnostic communication and stewardship support.

It covers:

- rapid molecular result communication;
- microbiology and organism-identification workflow considerations;
- AST status and follow-up communication;
- *C. auris* suspicion, identification, screening, and transfer-communication awareness;
- LIS alert/comment language examples;
- reflex and confirmatory testing considerations;
- TAT and escalation metrics; and
- staff training concepts.

It is designed for public review and adaptation. It uses synthetic examples and public guidance, not patient records or employer-controlled procedures.

Non-Scope

This package does not:

- prescribe antimicrobial or antifungal therapy;
- recommend a specific drug, dosage, or treatment duration;
- direct patient-specific clinical decisions;
- replace clinical judgment;
- replace institutional policy;
- replace laboratory director approval;
- replace CLIA, CAP, state, local, or institutional requirements;
- establish a public-health reporting system;

- claim adoption by any employer, hospital, public-health agency, or professional body; or
- use confidential employer information or patient data.

Design Principles

1. Laboratory-Scoped

The artifacts focus on what laboratory personnel can reasonably document, verify, communicate, and escalate under local policy. The package does not cross into prescribing, diagnosis ownership, infection-control orders, or treatment selection.

2. Non-Confidential

The examples are synthetic. They should not contain patient names, medical record numbers, accession numbers, dates of service, facility identifiers, local screenshots, internal SOP language, or employer data.

3. Public And Portable

The package is meant to be readable outside one institution. It uses generic workflow categories and neutral wording so other laboratories can evaluate and adapt the framework within their own governance structures.

4. Confirmation-Aware

Rapid molecular or organism-identification findings should be communicated with attention to result status. Preliminary findings, pending culture or AST, confirmatory testing, corrected results, and final reports may require different comments, routing, and follow-up.

5. Measurement-Aware

The package includes a practical metric plan so a laboratory can track whether communication steps are happening consistently. Example measures include time from verified result to documented notification, final-result follow-up completion, and staff training completion.

6. Locally Governed

Any implementation must be reviewed and approved by the relevant local authorities, such as the laboratory director, medical director, infection prevention, antimicrobial stewardship, quality, information technology, and compliance teams.

The public package may help organize that review, but it does not itself authorize a laboratory to change live LIS comments, notification routes, report wording, or clinical workflows.

Public Source Basis

This package is informed by public guidance from the Centers for Disease Control and Prevention.

Relevant source themes include:

- Diagnostic excellence and diagnostic stewardship emphasize appropriate test use, interpretation, communication, multidisciplinary teamwork, and systems that support timely diagnosis.
- Antibiotic stewardship programs promote appropriate antibiotic use and efforts to combat antimicrobial resistance.
- MDRO containment guidance addresses response to novel and targeted MDROs in healthcare facilities and public-health settings.
- *C. auris* guidance emphasizes healthcare transmission risk, staff education, infection-control communication, and communication of *C. auris* status during transfer or discharge.
- CDC laboratory guidance for *C. auris* describes the importance of early accurate identification, reporting, MALDI-TOF, PCR, and laboratory communication with infection prevention/public-health channels.

These sources provide background. The specific contribution of this reference package is the laboratory-facing communication workflow and accompanying artifacts.

Artifact Roadmap

This technical note anchors a six-part package.

Artifact 1: Public Technical Note

Defines the package, scope, source basis, boundaries, intended audience, and local adaptation model.

Artifact 2: Rapid MDRO Result-To-Action Decision Tree

Provides a high-level workflow from rapid MDRO-related trigger to laboratory verification, communication routing, reflex/confirmatory review, and documentation.

Artifact 3: Reflex And Confirmatory Testing Matrix

Maps common rapid MDRO-related trigger types to preliminary/final/corrected result status, suggested laboratory checks, possible reflex or confirmatory paths, communication actions, and caveats.

Artifact 4: LIS Alert/Comment Template Bank

Provides neutral, adaptable LIS comment examples for preliminary rapid results, resistance-marker detection, organism identification pending confirmation, AST pending/finalized status, *C. auris* concerns, interfacility transfer communication reminders, and corrected/final result follow-up.

Artifact 5: TAT And Escalation Metric Plan

Defines practical metrics for tracking communication consistency, result-to-notification timing, final-result follow-up, training completion, and artifact feedback incorporation.

Artifact 6: Staff Training Checklist

Provides a checklist and synthetic case exercises for laboratory staff reviewing rapid MDRO communication workflows.

Local Adaptation Model

Laboratories reviewing this package should adapt it through local governance. A safe adaptation process may include:

1. Laboratory director or medical director review.
2. Quality and compliance review.
3. Infection prevention and antimicrobial stewardship review where applicable.
4. LIS/informatics review for any local comment or alert changes.
5. Staff training and competency documentation.
6. Pilot review using synthetic or approved non-patient examples before any live workflow change.
7. Ongoing monitoring of communication and follow-up metrics.

This model is intentionally modular. A laboratory may adapt the LIS comment templates without using the full metric plan, or may use the training checklist to review existing local procedures.

Any local adaptation should replace the synthetic examples with locally approved training or implementation materials only through approved systems and privacy/compliance review.

Example Use Case

The following synthetic scenario illustrates the package boundary.

Scenario:

A rapid molecular blood culture panel detects an organism and a resistance marker. Culture and full AST are still pending.

Laboratory workflow supported by this package:

1. Confirm result status and internal review/QC completion.
2. Use locally approved notification route if the result meets urgent or critical communication criteria.
3. Add neutral LIS comment language that notes rapid detection and pending culture/AST status.
4. Route infection prevention or stewardship awareness according to institutional policy.
5. Track whether final AST or corrected results require follow-up communication.

Out of scope:

The laboratory artifact does not select antibiotic therapy, recommend a medication, or override institutional protocols.

Review Questions For Technical Reviewers

Reviewers may evaluate this package by asking:

- Is the scope appropriately limited to laboratory communication and diagnostic stewardship support?
- Are the examples clear enough for laboratory professionals to understand?
- Do the workflows respect preliminary, final, corrected, and pending-result distinctions?
- Are the LIS comment examples neutral and non-prescriptive?
- Could the artifacts be adapted by more than one high-complexity laboratory?
- Are any claims too broad or unsupported?
- Are there additional safety boundaries that should be added?

References

- CDC, Core Elements of Hospital Diagnostic Excellence Programs: <https://www.cdc.gov/patient-safety/hcp/hospital-dx-excellence/index.html>
- CDC, Core Elements of Hospital Antibiotic Stewardship Programs: <https://www.cdc.gov/antibiotic-use/hcp/core-elements/hospital.html>
- CDC, MDRO Containment Strategy: <https://www.cdc.gov/healthcare-associated-infections/php/preventing-mdros/mdro-containment-strategy.html>
- CDC, Infection Control Guidance: Candida auris: <https://www.cdc.gov/candida-auris/hcp/infection-control/index.html>
- CDC, Laboratory Information for C. auris: <https://www.cdc.gov/candida-auris/hcp/laboratories/index.html>
- CDC, Identification of C. auris: <https://www.cdc.gov/candida-auris/hcp/laboratories/identification-of-c-auris.html>
- CDC, Guidance for Detection of C. auris Colonization: <https://www.cdc.gov/candida-auris/hcp/laboratories/detection-colonization.html>

Rapid MDRO Result-To-Action Decision Tree

Title: Laboratory Communication Workflow for Rapid MDRO-Related Results

Author: Marios Akugri

Version: 0.4 reviewer-ready draft

Date: 2026-08-25

Summary

This decision tree provides a laboratory-scoped workflow for handling rapid MDRO-related result signals. It is designed to help laboratory personnel organize result verification, communication routing, reflex or confirmatory review, LIS comment use, documentation, and final-result follow-up.

This is not a clinical treatment decision tree. It does not prescribe therapy, direct patient management, or replace local institutional protocols.

Intended Use

Use this workflow as a public reference example for:

- rapid molecular panel results;
- organism identification findings;
- resistance-marker detection;
- antimicrobial susceptibility testing status;
- suspected or confirmed *C. auris* findings;
- corrected or final result follow-up; and
- communication/documentation consistency.

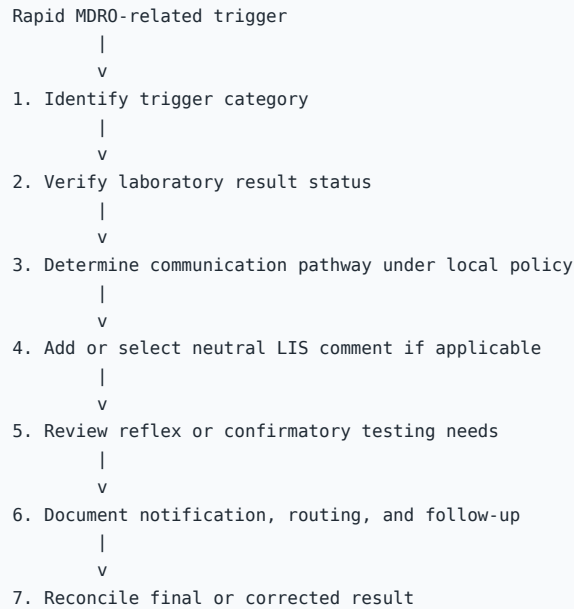
Any local implementation would require review by the laboratory director, medical director, infection prevention, antimicrobial stewardship, quality/compliance, and LIS/informatics stakeholders as applicable.

Evidence-Safe Boundary

This artifact uses generic workflow categories and synthetic examples only. It does not include patient data, accession numbers, local facility data, employer SOP language, or confidential LIS screenshots.

This artifact supports workflow review only. It does not show clinical assay validation, live LIS configuration, employer adoption, or public-health reporting implementation.

Core Workflow



The workflow is cyclic. If a preliminary result later becomes final, if AST results become available, or if a corrected report changes earlier understanding, the laboratory can re-enter the workflow at Step 2 and determine whether follow-up communication is needed under local policy.

Step 1: Identify Trigger Category

Trigger Category	Synthetic Example	Laboratory Question
Rapid molecular detection	Blood culture molecular panel detects organism and resistance marker.	Is this a preliminary rapid result with culture/AST pending?
Organism-identification concern	Yeast identification suggests possible <i>C. auris</i> or related organism requiring confirmation.	Does the organism require additional identification or referral under local policy?
Resistance phenotype	AST suggests resistant phenotype relevant to MDRO communication.	Is the AST preliminary, final, or corrected?
Colonization screening	Screening result suggests <i>C. auris</i> colonization.	Is infection prevention notification or transfer-status communication required by local policy?
Discrepant result	Rapid result and culture/AST do not fully align.	Does the discrepancy require corrected reporting, comment update, or technical review?
Corrected/final result	Final AST or corrected organism ID changes the earlier interpretation.	Is follow-up communication needed?

Step 2: Verify Laboratory Result Status

Before communication, confirm the result status.

Status Check	Purpose
Preliminary, final, or corrected status	Prevents a rapid result from being communicated as more definitive than it is.
QC/review completion	Confirms the result has passed required laboratory review steps.
Specimen/result context	Confirms the result belongs to the correct specimen type and testing pathway.
Culture/AST pending status	Supports clear comment language and follow-up planning.
Local critical/urgent communication category	Determines whether the result meets institutional notification requirements.

If any required internal review is incomplete, the laboratory should follow local policy before releasing, commenting, or escalating.

Step 3: Determine Communication Pathway

This workflow does not define mandatory recipients. It helps laboratories identify possible communication pathways that may exist under local policy.

Communication Pathway	When It May Be Relevant	Boundary
Clinical team/provider notification	Result meets local urgent, critical, or rapid-result communication criteria.	Laboratory communicates result status; clinical team manages patient care.
Infection prevention notification	Finding may affect isolation, transfer communication, or facility containment workflow.	Laboratory routes awareness under local protocol; infection prevention determines local action.
Antimicrobial stewardship/pharmacy awareness	Rapid organism/resistance information may be reviewed by a stewardship team under local process.	Laboratory does not recommend therapy.
Laboratory supervisor/technical review	Result is discrepant, corrected, unusual, or requires confirmatory pathway review.	Technical review follows laboratory policy.
Public-health reporting route	Result falls under jurisdictional or institutional reporting requirements.	Reporting follows local/state/federal rules and institutional policy.

Step 4: Select Neutral LIS Comment If Applicable

LIS comments should be neutral, status-aware, and non-prescriptive.

Recommended style:

Rapid molecular testing detected [organism/resistance marker]. Culture and/or susceptibility testing may be pending. Correlate with final laboratory results and institutional notification protocols.

Avoid language that:

- selects therapy;
- tells a provider to start or stop a drug;
- suggests a patient-specific management decision;
- overstates preliminary results as final; or
- claims a local protocol exists where none has been approved.

Step 5: Review Reflex Or Confirmatory Testing Needs

Rapid MDRO-related signals often require status-aware follow-up.

Result Situation	Possible Laboratory Follow-Up
Rapid molecular result positive, culture pending	Track final culture and AST status.
Resistance marker detected, phenotype pending	Track AST availability and final reporting.
Organism identification suggests <i>C. auris</i>	Review local confirmatory identification pathway and reporting route.
<i>C. auris</i> colonization screen positive	Confirm local communication and infection-prevention routing.
Preliminary result later corrected	Determine whether revised communication and LIS comment update are needed.
Rapid result negative, culture still pending	Avoid closing the communication loop until final reporting rules are satisfied.

This step does not mandate testing. It prompts review under local validation, laboratory director approval, and institutional procedures.

Step 6: Document Notification, Routing, And Follow-Up

For any result that triggers a communication pathway, documentation should be complete enough for quality review.

Suggested documentation elements:

- result trigger category;
- result status at time of communication;
- communication route used;
- date/time of documented notification, if applicable;
- recipient role or approved routing queue, if applicable;
- LIS comment used, if applicable;
- pending culture/AST/final-result follow-up flag;
- corrected result follow-up, if applicable; and
- staff initials or authenticated laboratory documentation per local policy.

Do not create duplicate documentation outside approved systems if local policy does not allow it.

Step 7: Reconcile Final Or Corrected Result

The workflow is not complete when a rapid preliminary result is communicated. It should include a final-result check.

Final-result reconciliation should ask:

- Did culture confirm the rapid result?
- Did AST become available after the rapid result?
- Did the organism identification change?
- Did a corrected report issue?
- Does the LIS comment need updating under local policy?
- Does infection prevention, stewardship, the clinical team, or laboratory leadership require follow-up under local policy?
- Was follow-up documented?

Synthetic Walkthrough

Trigger:

Rapid molecular panel detects Organism X and Resistance Marker Y from a positive blood culture bottle.

Step 1:

Trigger category = rapid molecular detection with resistance-marker detection.

Step 2:

Result status = preliminary rapid result. Culture and AST pending.

Step 3:

Local policy determines whether clinical team, infection prevention, or stewardship notification applies.

Step 4:

Neutral LIS comment notes rapid detection and pending culture/AST status.

Step 5:

Culture and AST follow-up flag is created.

Step 6:

Notification route and comment use are documented according to local policy.

Step 7:

When final culture/AST posts, the laboratory reviews whether follow-up communication or comment update is required.

Out of scope: this walkthrough does not select a medication, determine isolation orders, or replace local policy.

Quality Review Questions

A laboratory adapting this workflow may ask:

- Are trigger categories clearly defined?
- Does the workflow distinguish preliminary, final, and corrected results?
- Is the communication route approved locally?
- Are LIS comments neutral and non-prescriptive?
- Is reflex or confirmatory testing language consistent with local validation and regulatory requirements?
- Does the workflow include final-result follow-up?
- Can communication documentation be audited?

Safety Notes

- Use only locally approved result-reporting and notification pathways.
- Do not use this artifact as a standalone SOP.
- Do not enter sample comments into a live LIS without institutional review and approval.
- Do not use patient information to test or demonstrate this workflow outside approved systems.
- Do not represent this artifact as adopted by any institution unless documented.

Related Suite Artifacts

- `01_public_technical_note.md`
- `03_reflex_confirmatory_testing_matrix.md`
- `04_lis_alert_comment_templates.md`
- `05_tat_escalation_metric_plan.md`
- `06_staff_training_checklist.md`

Reflex And Confirmatory Testing Matrix

Title: Laboratory Matrix For Rapid MDRO Result Status, Confirmation, Communication, And Follow-Up

Author: Marious Akugri

Version: 0.4 reviewer-ready draft

Date: 2026-08-25

Summary

This matrix provides synthetic, laboratory-scoped examples for handling rapid MDRO-related result signals. It is designed to help laboratories distinguish preliminary, final, corrected, pending, and confirmation-dependent findings before communication or follow-up.

This artifact is not a testing menu, treatment guide, institutional policy, or public-health reporting directive. It is a reference framework that laboratories may adapt under local validation, regulatory, medical director, laboratory director, infection prevention, antimicrobial stewardship, and quality oversight.

The matrix uses "confirmatory" and "validation" language in the ordinary laboratory-review sense. It does not claim that Marious Akugri has clinically validated a test method, implemented this matrix in a hospital, or obtained institutional approval for any live workflow.

How To Use This Matrix

Use the matrix to ask four laboratory questions:

1. What type of result or trigger is present?
2. What is the result status?
3. What laboratory follow-up may be needed?
4. What communication or documentation step may be appropriate under local policy?

All examples are synthetic. Local policies, approved methods, validation/verification records, and institutional governance control actual testing and communication.

Matrix

Trigger/Result Type	Synthetic Example	Result Status	Suggested Laboratory Check	Possible Reflex/Confirmatory Path	Communication Action	Scope of Application
Rapid molecular blood culture panel positive	Panel detects Organism X and Resistance Marker Y from a positive blood culture bottle.	Preliminary rapid result; culture and AST pending.	Confirm internal review/QC status and whether result is reportable under local policy.	Track culture growth, organism confirmation, and AST finalization.	Use approved notification route if result meets local urgent/critical communication policy; add neutral LIS comment noting pending culture/AST.	Do not select therapy. Communicate to laboratory only.
Rapid molecular resistance marker detected	Resistance Marker Y detected, but phenotypic susceptibility is pending.	Preliminary molecular signal.	Confirm marker/result association and result-release rules.	Follow local AST and organism-confirmation workflow.	Notify/reroute according to local policy; note that AST may be pending.	Avoid implying phenotypic resistance has been finalized if AST is pending.

Trigger/Result Type	Synthetic Example	Result Status	Suggested Laboratory Check	Possible Reflex/Confirmatory Path	Communication Action	Scope of Action
Organism identification concern	Yeast isolate identified by a method with known <i>C. auris</i> misidentification limitations.	Identification concern; may require confirmation.	Review identification method and whether further characterization is required.	Confirm with locally validated method, reference lab, MALDI-TOF database, sequencing, or other approved pathway.	Escalate to laboratory supervisor/director or infection prevention route according to local policy.	Do not report speculative organism ID as confirmed
Suspected <i>C. auris</i>	Isolate or screen result raises <i>C. auris</i> concern.	Suspected, preliminary, or confirmed depending on method.	Confirm method, validation status, specimen type, and required reporting/notification rules.	Follow local <i>C. auris</i> confirmation and public-health communication pathway.	Communicate <i>C. auris</i> concern/status under local policy, including transfer-status awareness if applicable.	Do not claim infection vs colonization the result type local interpretation
<i>C. auris</i> colonization screening positive	Skin swab screening result positive in a synthetic example.	Screening result; colonization context.	Confirm whether the screening method is validated and whether result is final.	Follow local infection prevention and reporting pathway.	Notify infection prevention or designated route according to local policy.	CDC notes <i>C. auris</i> status should communicate during transfer/disclosure local policy process.
Resistant AST phenotype	Final AST shows resistant phenotype relevant to MDRO workflow.	Final AST result.	Confirm AST method, result review, and whether any critical/urgent reporting rule applies.	If needed, verify unusual or unexpected phenotype according to local policy.	Communicate final AST status using approved result-release and notification process.	Do not record antibiotic sensitivity
Preliminary AST or automated result pending verification	Instrument produces preliminary susceptibility pattern requiring review.	Preliminary or unverified.	Confirm result review, QC, flags, organism context, and final-release criteria.	Repeat, confirm, suppress, or finalize only under local policy and supervisory review.	Avoid final communication until local release criteria are met, unless local policy allows preliminary communication.	Do not over-report unverified results
Discrepant rapid and culture result	Rapid signal suggests organism/resistance marker, but culture/AST later does not align.	Discrepant; follow-up needed.	Review specimen, instrument flags, possible mixed culture, contamination, or technical issue under local policy.	Technical review, repeat testing, corrected comment, or reference pathway as approved locally.	Notify laboratory supervisor/technical review route; follow local corrected-result policy if needed.	Do not hide discrepancies document follow-up.
Corrected organism identification	Initial ID reported, later corrected after additional method.	Corrected result.	Confirm corrected report approval and affected comments/notifications.	Update LIS/result record according to local policy.	Determine whether prior recipients require follow-up communication.	Corrected results require careful documentation
Final result after preliminary rapid result	Culture and AST finalize after earlier rapid molecular communication.	Final result available.	Compare final result to prior rapid communication.	Close follow-up flag if no discrepancy; escalate if discrepancy or major status change exists.	Document final-result reconciliation and any needed follow-up.	Rapid workflow should not end before final-review.
Negative rapid panel with culture pending	Rapid panel negative, but culture workup continues.	Negative rapid result; culture pending.	Confirm panel scope and what organisms/markers are not covered.	Continue routine culture/AST workflow as applicable.	Avoid over-reassurance; use neutral wording if commenting.	Negative rapid result may not exclude all organisms or resistance mechanisms

Trigger/Result Type	Synthetic Example	Result Status	Suggested Laboratory Check	Possible Reflex/Confirmatory Path	Communication Action	Scope of Impact
Transfer communication reminder	Known C. auris or targeted MDRO status relevant to transfer documentation.	Prior or current status depending on record.	Confirm result history, current status, and local transfer communication policy.	Follow institutional transfer communication process.	Notify receiving facility/unit route if local policy requires.	Decisions to transfer/accept clinical/institutional laboratory result status communication support.

Status Language Controls

Use status-aware language.

If The Result Is...	Safer Comment Logic
Preliminary	State that the rapid/preliminary result was detected and that culture/AST/final identification may be pending.
Final	State final laboratory result status without treatment recommendation.
Corrected	State that a corrected result has been issued and follow local corrected-result notification policy.
Pending confirmation	State that additional confirmation or characterization may be pending under local policy.
Discrepant	Route for technical review and follow local discrepancy/corrected-reporting process.

Communication Priority Categories

This matrix does not define mandatory priority levels. A laboratory can adapt categories such as:

Category	Description	Example Documentation Need
Routine result follow-up	Result does not meet urgent criteria but requires final-status tracking.	Final review complete, comment applied if applicable.
Urgent/critical notification	Result meets local urgent or critical communication policy.	Recipient, time, route, and staff documentation.
Infection prevention awareness	Result may affect isolation, containment, or transfer communication.	Route or queue used, time, and final-status follow-up.
Stewardship awareness	Result may be reviewed by antimicrobial stewardship under local process.	Route used and result status noted.
Technical review	Result is unusual, discrepant, corrected, or confirmation-dependent.	Supervisor/director review and follow-up action.

Synthetic Example: C. auris Identification Concern

Trigger:

A yeast isolate is identified by a method that may have C. auris misidentification limitations.

Status:

Identification concern, not confirmed C. auris.

Laboratory check:

Review identification method, result confidence, organism context, and local confirmation pathway.

Possible follow-up:

Use approved confirmatory method or referral route if local criteria are met.

Communication:

Route to laboratory supervisor/director and infection prevention according to local policy.

Comment boundary:

Do not label the isolate as confirmed C. auris until confirmation criteria are satisfied.

Synthetic Example: Rapid Resistance Marker With AST Pending

Trigger:

Rapid molecular testing detects a resistance marker from a positive blood culture bottle.

Status:

Preliminary rapid molecular result. AST pending.

Laboratory check:

Confirm result review/QC status and reporting criteria.

Possible follow-up:

Track final organism identification and AST.

Communication:

Use local urgent/critical notification route if applicable. LIS comment should state that culture and/or AST may be pending.

Comment boundary:

Do not provide treatment selection or imply final phenotypic susceptibility before AST is finalized.

Quality Review Checklist

Before adapting this matrix, check:

- Does each row distinguish result status?
- Is every communication action tied to local policy?
- Are preliminary and final results clearly separated?
- Does every confirmation-dependent result avoid overstatement?
- Are public-health reporting references limited to local/state/federal rules and institutional process?
- Are clinical treatment decisions excluded?
- Are all examples synthetic?

Related Suite Artifacts

- 01_public_technical_note.md
- 02_rapid_mdرو_result_to_action_decision_tree.md
- 04_lis_alert_comment_templates.md
- 05_tat_escalation_metric_plan.md

- 06_staff_training_checklist.md

LIS Alert And Comment Template Bank

Title: Neutral Laboratory Comment Examples For Rapid MDRO-Related Results

Author: Marious Akugri

Version: 0.4 reviewer-ready draft

Date: 2026-08-25

Summary

This template bank provides neutral, synthetic examples of LIS alert and result-comment wording for rapid MDRO-related laboratory findings. The examples are designed to help laboratories communicate result status, pending confirmation, pending AST, infection-prevention awareness, stewardship awareness, and final-result follow-up without directing clinical treatment.

These examples are not approved institutional comments. They must not be inserted into a live LIS unless reviewed and approved through local laboratory, medical, infection prevention, stewardship, quality, compliance, and LIS governance.

This template bank is for drafting and review. It does not represent a live LIS build, employer-approved comment bank, or clinical validation record.

General Comment Formula

Use a three-part structure:

[Result status]. [Laboratory limitation or pending follow-up]. [Local-policy routing or correlation statement].

Example:

Rapid molecular testing detected [organism/resistance marker]. Culture and/or susceptibility testing may be pending. Correlate with final laboratory results and institutional notification protocols.

Wording Controls

Use language that:

- identifies the laboratory finding;
- states whether the result is preliminary, final, corrected, or pending confirmation;
- notes whether culture, organism identification, AST, or confirmatory testing may still be pending;
- routes awareness according to local policy; and
- avoids patient-specific treatment direction.

Avoid language that:

- selects an antimicrobial or antifungal agent;
- states a drug should be started, stopped, changed, or avoided;
- directs isolation or transfer decisions;
- claims final confirmation when confirmation is pending;
- bypasses local institutional policy; or
- claims adoption by any institution.

Placeholder Terms

Use placeholders until locally approved language is created:

Placeholder	Meaning
[organism]	Organism detected or identified.
[resistance marker]	Molecular resistance marker or resistance-related finding.
[specimen type]	General specimen category. Do not include patient identifiers.
[method]	Locally approved or validated method, if appropriate for comment use.
[local policy]	Institution-specific protocol or notification pathway.
[receiving team]	Locally approved team or route, such as provider, infection prevention, or stewardship queue.

Template 1: Preliminary Rapid Molecular Detection

Use when a rapid molecular result is available before full culture or susceptibility follow-up.

Rapid molecular testing detected [organism] from [specimen type]. Culture and/or susceptibility testing may be pending. Correlate with final laboratory results and institutional notification protocols.

Short version:

Preliminary rapid molecular result: [organism] detected. Final culture and/or susceptibility results may be pending.

Scope note:

This comment communicates laboratory status only. It does not direct clinical treatment.

Template 2: Resistance Marker Detected, AST Pending

Use when a molecular resistance marker is detected but phenotypic AST is not final.

Rapid molecular testing detected [resistance marker]. Phenotypic susceptibility testing may be pending. Correlate with final AST and institutional notification protocols.

Alternative:

[resistance marker] detected by rapid molecular testing. Final organism identification and/or susceptibility testing may be pending.

Scope note:

Do not imply that final phenotypic susceptibility has been completed if AST is pending.

Template 3: Organism Identification Pending Confirmation

Use when organism identification may require confirmation or further characterization.

Preliminary organism identification suggests [organism or organism group]. Additional confirmation or characterization may be pending according to laboratory protocol.

Alternative:

Organism identification is preliminary and may require confirmatory testing. Interpret with final laboratory report and local notification protocol.

Scope note:

Do not label an organism as confirmed if local confirmation criteria have not been met.

Template 4: C. auris Concern Pending Confirmation

Use when C. auris is suspected or confirmation is pending.

Preliminary laboratory finding raises concern for Candida auris. Confirmatory identification and institutional notification steps may be pending according to local protocol.

Alternative:

Candida auris confirmation status: pending. Follow local laboratory and infection prevention notification procedures.

Scope note:

This comment should not be used to imply confirmed C. auris unless confirmation criteria are satisfied.

Template 5: Confirmed C. auris Laboratory Result

Use only when local confirmation and reporting criteria are satisfied.

Candida auris identified by [method]. Follow institutional infection prevention, communication, and reporting protocols as applicable.

Alternative:

Confirmed Candida auris laboratory result. Communicate status according to local infection prevention and transfer communication procedures.

Scope note:

The comment communicates laboratory status. Infection prevention actions and patient-care decisions remain governed by local policy.

Template 6: AST Pending

Use when an organism result is available but susceptibility testing is not final.

[organism] identified. Antimicrobial susceptibility testing is pending. Correlate with final AST report when available.

Alternative:

Organism identification available; susceptibility testing not yet final.

Scope note:

Do not infer susceptibility or resistance beyond the reported result status.

Template 7: AST Finalized

Use when susceptibility testing is final and local result release is complete.

Final antimicrobial susceptibility testing is available for [organism]. Interpret according to the final laboratory report and institutional clinical protocols.

Alternative:

Final AST result posted for [organism]. Review final laboratory report for susceptibility interpretation.

Scope note:

The comment points to the final laboratory report. It does not choose treatment.

Template 8: Corrected Result Follow-Up

Use when a corrected report changes a prior organism, resistance-marker, AST, or status communication.

Corrected laboratory result issued for [organism/result category]. Prior preliminary or final communication may require review according to local corrected-result notification policy.

Alternative:

Corrected report: [result category] updated. Follow institutional corrected-result communication procedure.

Scope note:

Corrected result comments should follow local policy and be documented through approved systems.

Template 9: Discrepant Rapid And Culture/AST Result

Use when a rapid finding and later culture/AST workup do not align.

Rapid result and follow-up culture and/or susceptibility findings require technical review. Interpret according to final corrected laboratory report and local documentation protocol.

Alternative:

Result discrepancy identified between preliminary rapid testing and follow-up laboratory findings. Technical review and final reporting are pending.

Scope note:

Do not suppress discrepancies without technical review under local policy.

Template 10: Interfacility Transfer Communication Reminder

Use only where local policy supports transfer communication reminders.

MDRO or Candida auris status may require communication during patient transfer or discharge according to institutional policy.

Alternative:

If transfer or discharge occurs, communicate applicable MDRO/Candida auris status according to local infection prevention and transfer communication procedures.

Scope note:

This comment does not determine transfer eligibility, bed placement, isolation orders, or patient management.

Template 11: Infection Prevention Awareness

Use where local policy includes infection prevention routing for specified findings.

This laboratory result may meet institutional criteria for infection prevention awareness. Route according to local notification procedure.

Alternative:

Infection prevention notification may be indicated under local policy for this result category.

Scope note:

The laboratory routes awareness; infection prevention determines local action.

Template 12: Antimicrobial Stewardship Awareness

Use where local policy includes stewardship routing for specified findings.

This result category may be reviewed by antimicrobial stewardship according to institutional process. Correlate with final organism identification and AST status.

Alternative:

Stewardship awareness route may apply under local policy. Final culture and/or AST status should be reviewed when available.

Scope note:

The laboratory does not recommend medication selection.

Comment Review Checklist

Before a comment is adapted, ask:

- Does it state result status clearly?
- Does it avoid therapy recommendations?
- Does it avoid isolation or transfer orders?
- Does it distinguish preliminary, final, corrected, and pending results?
- Does it avoid overstating rapid molecular findings?
- Does it route awareness only under local policy?
- Has LIS/informatics reviewed the wording and display behavior?
- Has laboratory/medical leadership approved the comment?

Related Suite Artifacts

- 01_public_technical_note.md
- 02_rapid_mdro_result_to_action_decision_tree.md
- 03_reflex_confirmatory_testing_matrix.md
- 05_tat_escalation_metric_plan.md
- 06_staff_training_checklist.md

TAT And Escalation Metric Plan

Title: Measurement Plan For Rapid MDRO Laboratory Communication Workflows
Author: Marious Akugri
Version: 0.4 reviewer-ready draft
Date: 2026-08-25

Summary

This metric plan defines non-confidential ways to evaluate a rapid MDRO diagnostic communication workflow. It focuses on laboratory process measures: turnaround time, result verification, notification documentation, final-result follow-up, corrected-result follow-up, and staff training completion.

The plan uses synthetic examples only. It does not report real employer performance, patient outcomes, institutional quality data, or protected health information.

For reviewer circulation, these metrics are proposed process measures. They are not presented as live baseline data, quality-improvement results, or proof that any institution has adopted the workflow.

Purpose

The purpose of this metric plan is to help laboratories evaluate whether rapid MDRO-related results are handled consistently after detection.

The plan asks:

- Was the rapid result verified?
- Was the result status clear?
- Was the appropriate communication route used under local policy?
- Was the notification documented?
- Was culture/AST/final-result follow-up completed?
- Was corrected-result follow-up completed when needed?
- Were staff trained on the workflow?

Measurement Boundary

This metric plan measures laboratory workflow consistency. It does not measure:

- medication choice;
- clinical treatment decisions;
- provider response;
- patient outcomes;
- infection rates;
- mortality;
- hospital length of stay; or
- public-health reporting completeness.

Those areas may be studied by qualified institutional teams under appropriate approvals, but they are outside this public reference artifact.

Metric Categories

Metric Category	Question Answered	Example Use
Result-to-verification time	How long from result availability to laboratory verification?	Identifies delays in internal review.

Metric Category	Question Answered	Example Use
Verification-to-notification time	How long from verified result to documented communication?	Measures communication workflow reliability.
Final-result follow-up completion	Were pending culture/AST/final results reconciled?	Prevents rapid preliminary results from becoming orphaned.
Corrected-result follow-up	Were corrected reports routed or documented?	Supports quality and patient-safety communication.
LIS comment consistency	Was the correct approved comment category used?	Supports standardized result interpretation language.
Training completion	Did relevant staff review the workflow?	Supports local adaptation readiness.
Reviewer feedback incorporation	Were technical review comments tracked and addressed?	Supports continuous improvement before wider dissemination.

Core Metrics

Metric 1: Result-To-Verification Time

Definition:

Time from rapid MDRO-related result availability to laboratory verification/documented review.

Example fields:

- synthetic event ID;
- result availability timestamp;
- verification timestamp;
- result status at verification;
- verifier role/category.

Use:

- identify whether rapid signals are reviewed promptly;
- compare workflow before and after staff training in a synthetic or approved test environment.

Metric 2: Verification-To-Notification Time

Definition:

Time from laboratory verification to documented notification or routing under local policy.

Example fields:

- synthetic event ID;
- verification timestamp;
- communication route;
- notification timestamp;
- recipient role/category or routing queue;
- documentation completed: yes/no.

Use:

- evaluate whether approved communication routes are followed consistently.

Metric 3: Final-Result Follow-Up Completion

Definition:

Percentage of rapid preliminary MDRO-related events with documented review of culture, AST, final identification, or final report status.

Example formula:

$$\text{Final-result follow-up completion} = \frac{\text{events with documented final-result review}}{\text{rapid preliminary events requiring follow-up}}$$

Use:

- ensure preliminary rapid results are reconciled after final results are available.

Metric 4: Corrected-Result Follow-Up Completion

Definition:

Percentage of corrected MDRO-related reports with documented communication review under local policy.

Example formula:

$$\text{Corrected-result follow-up completion} = \frac{\text{corrected events with documented follow-up}}{\text{corrected events requiring follow-up}}$$

Use:

- support quality review where corrected results change prior communication status.

Metric 5: LIS Comment Consistency

Definition:

Percentage of applicable rapid MDRO-related events using the locally approved comment category for the result status.

Example formula:

$$\text{LIS comment consistency} = \frac{\text{events with correct comment category}}{\text{events where a comment was applicable}}$$

Use:

- evaluate whether preliminary, pending, final, corrected, and confirmation-dependent comments are used correctly.

Metric 6: Staff Training Completion

Definition:

Percentage of relevant staff who complete workflow review and synthetic case exercises.

Example formula:

$$\text{Training completion} = \frac{\text{staff with completed checklist}}{\text{staff assigned to workflow review}}$$

Use:

- document dissemination and readiness before any local adaptation.

Metric 7: Reviewer Feedback Incorporation

Definition:

Percentage of technical review comments triaged and addressed in the artifact suite.

Example formula:

Feedback incorporation =
review comments closed / total review comments received

Use:

- show that the public artifact suite improves through outside technical review.

Synthetic Tracking Table

Use synthetic IDs and generic timestamps for training or demonstration.

Event ID	Trigger Type	Result Status	Verification Complete	Communication Route	Notification Documented	Final Follow-Up Needed	Final Follow-Up Complete	Comment Category
SIM-001	Rapid molecular + resistance marker	Preliminary	Yes	Clinical team route	Yes	Yes	Pending	Resistance marker, AST pending
SIM-002	Suspected C. auris	Pending confirmation	Yes	Infection prevention route	Yes	Yes	Pending	C. auris pending confirmation
SIM-003	AST resistant phenotype	Final	Yes	Local notification route	Yes	No	N/A	Final AST available
SIM-004	Rapid negative, culture pending	Preliminary	Yes	Routine result follow-up	N/A	Yes	Pending	Negative rapid, culture pending
SIM-005	Corrected organism ID	Corrected	Yes	Corrected-result route	Yes	Yes	Yes	Corrected result follow-up

This table is illustrative only. It is not derived from any patient or employer data.

Example Dashboard Fields

A laboratory adapting this workflow could track:

- total rapid MDRO-related trigger events reviewed;
- percentage with verified result status;
- median verification-to-notification time;
- percentage with documented notification where required;
- percentage with final-result follow-up complete;
- percentage with corrected-result follow-up complete;
- percentage with appropriate comment category;
- training completion rate; and
- open reviewer feedback items.

Review Cadence

Suggested review cadence for a local adaptation:

Stage	Review Activity
Pre-use	Review workflow against local policy using synthetic examples.
Training	Staff complete checklist and synthetic case exercises.

Stage	Review Activity
Limited local review	Quality/lab leadership reviews documentation behavior in an approved setting.
Ongoing review	Review metrics monthly or according to local quality calendar.
Revision	Update comments, triggers, or training materials based on approved feedback.

Safety And Compliance Notes

- Do not export patient data to build a public example.
- Do not use live data outside approved systems.
- Do not create unofficial spreadsheets containing patient identifiers.
- Do not report institutional metrics publicly without authorization.
- Do not claim this metric plan improves patient outcomes unless an approved study supports that claim.
- Use local policy for critical results, corrected results, infection prevention routing, stewardship routing, and public-health reporting.

Related Suite Artifacts

- 01_public_technical_note.md
- 02_rapid_mdرو_result_to_action_decision_tree.md
- 03_reflex_confirmatory_testing_matrix.md
- 04_lis_alert_comment_templates.md
- 06_staff_training_checklist.md

Staff Training Checklist

Title: Laboratory Staff Checklist For Rapid MDRO Result Communication Workflows

Author: Marious Akugri

Version: 0.4 reviewer-ready draft

Date: 2026-08-25

Summary

This checklist provides a synthetic training framework for laboratory staff reviewing rapid MDRO-related result communication workflows. It is intended to support consistent understanding of result status, verification, communication routing, LIS comment use, final-result follow-up, and documentation.

This checklist is not an institutional training record, competency assessment, clinical protocol, or treatment guide. Local laboratories must adapt and approve any training through their own governance processes.

Intended Learners

This checklist may be adapted for:

- medical laboratory scientists;
- clinical laboratory technologists;
- microbiology personnel;
- molecular diagnostics personnel;
- laboratory supervisors;
- quality personnel; and
- staff who participate in result review, release, notification, documentation, or final-result follow-up.

Training Objectives

After reviewing the workflow package, staff should be able to:

1. identify common rapid MDRO-related result triggers;
2. distinguish preliminary, final, corrected, pending, and confirmation-dependent result status;
3. recognize when culture, AST, organism identification, or confirmatory testing may still be pending;
4. apply neutral, non-prescriptive LIS comment logic;
5. follow local communication pathways for urgent, infection prevention, stewardship, or technical-review routing;
6. document notification and follow-up according to local policy;
7. reconcile rapid preliminary results with final or corrected results; and
8. understand what laboratory staff should not do when communicating rapid MDRO-related findings.

Training Boundary

This checklist does not train staff to:

- prescribe antimicrobial or antifungal therapy;
- select a drug or treatment plan;
- determine isolation orders;
- decide transfer acceptance or denial;
- replace infection prevention or antimicrobial stewardship judgment;
- bypass local laboratory director or medical director approval;
- create unapproved LIS comments; or
- use patient data in public examples.

Core Concepts Checklist

Staff should review and confirm understanding of each item.

Item	Reviewed	Notes
MDRO and AMR definitions.		
Difference between rapid molecular result, culture result, organism identification, and AST.		
Difference between preliminary, final, corrected, and pending result status.		
Local rules for result release and internal review/QC completion.		
Local urgent or critical result notification process.		
Local infection prevention notification route, if applicable.		
Local antimicrobial stewardship awareness route, if applicable.		
Local technical-review route for discrepant or unusual results.		
Local corrected-result communication procedure.		
Local public-health reporting route, if applicable.		
Approved LIS comment categories and display behavior.		
Final-result follow-up expectations after preliminary rapid results.		

What Laboratory Staff Should Do

Action	Purpose
Confirm result status before communication.	Avoid overstatement of preliminary or pending results.
Confirm internal review/QC completion.	Support reliable result release.
Use locally approved communication routes.	Keep communication consistent and auditable.
Use neutral LIS comments.	Communicate result status without directing treatment.
Track pending culture, AST, confirmation, final, or corrected results.	Prevent rapid preliminary results from becoming orphaned.
Document notification and follow-up according to local policy.	Support quality review and accountability.
Escalate unusual, discrepant, corrected, or confirmation-dependent results.	Ensure appropriate technical review.

What Laboratory Staff Should Not Do

Avoid	Reason
Do not select therapy.	Treatment decisions belong to clinical teams under institutional protocols.
Do not give drug-specific recommendations in LIS comments.	The lab comment should communicate laboratory status, not prescribe.
Do not state preliminary findings as final.	Preliminary rapid results may need culture, AST, or confirmatory follow-up.
Do not ignore corrected reports.	Corrected results may change prior communication status.
Do not use unapproved LIS comments.	Comments must be reviewed by local governance before live use.
Do not export patient identifiers into training examples.	Public or informal training examples must remain non-confidential.
Do not claim external adoption without documentation.	Public artifacts are reference materials unless adopted locally through approval.

Synthetic Case Exercise 1: Rapid Molecular Result With AST Pending

Scenario:

Rapid molecular testing detects Organism X and Resistance Marker Y from a positive blood culture bottle. Culture and AST are pending.

Staff questions:

1. Is the result preliminary, final, corrected, or pending confirmation?
2. What internal review/QC step must be complete before communication?
3. Which local communication route may apply?
4. What neutral LIS comment category fits this result?
5. What follow-up should be tracked when AST becomes available?

Expected training points:

- Do not imply final AST is complete.
- Use local urgent/critical route only if criteria are met.
- Document notification if required.
- Track final culture and AST follow-up.

Synthetic Case Exercise 2: Suspected *C. auris* Pending Confirmation

Scenario:

A yeast identification result raises concern for *Candida auris*, but confirmatory identification is pending.

Staff questions:

1. Can this be reported as confirmed *C. auris*?
2. What confirmatory or technical-review route may apply?
3. Which local infection prevention route may apply?
4. What LIS comment language avoids overstatement?
5. What should be documented?

Expected training points:

- Do not report confirmation before criteria are met.
- Escalate through local technical and infection prevention procedures.
- Use neutral pending-confirmation language.
- Follow local public-health reporting rules where applicable.

Synthetic Case Exercise 3: Corrected Result Follow-Up

Scenario:

A preliminary organism identification was communicated. Later, additional testing changes the organism identification and a corrected report is issued.

Staff questions:

1. What changed between the initial and corrected result?
2. Does local corrected-result communication policy apply?
3. Which prior recipients or routes may require follow-up?
4. Does an LIS comment or result note need updating?
5. How should follow-up be documented?

Expected training points:

- Corrected results require careful documentation.

- Prior communication may need follow-up.
- Follow local corrected-reporting procedure.

Synthetic Case Exercise 4: Negative Rapid Panel With Culture Pending

Scenario:

A rapid panel does not detect organisms or resistance markers included on the panel, but culture workup is still pending.

Staff questions:

1. What does the negative rapid result cover?
2. What remains pending?
3. What comment language avoids over-reassurance?
4. What final-result follow-up is needed?

Expected training points:

- Negative rapid results may not exclude all organisms or mechanisms.
- Culture/final result workflow should continue.
- Avoid language that closes the case prematurely.

Competency Confirmation Prompts

A local laboratory adapting this checklist may ask staff to confirm:

- I understand the difference between preliminary and final result status.
- I understand that LIS comments must not direct treatment.
- I understand that *C. auris* concern and confirmation require local technical and infection prevention procedures.
- I understand how to document notification according to local policy.
- I understand when final or corrected results require follow-up.
- I understand that synthetic examples must not be replaced with patient data outside approved systems.

Optional Training Record Template

This optional template is for local adaptation only.

Staff Role	Training Date	Workflow Reviewed	Synthetic Cases Completed	Questions Resolved	Reviewer/Trainer
MLS					
Molecular technologist					
Microbiology technologist					
Supervisor					

Do not use this table as an official training record unless adopted through local policy.

Related Suite Artifacts

- 01_public_technical_note.md
- 02_rapid_mdرو_result_to_action_decision_tree.md
- 03_reflex_confirmatory_testing_matrix.md
- 04_lis_alert_comment_templates.md

- 05_tat_escalation_metric_plan.md

Result-Status Taxonomy

Title: Common Vocabulary For Rapid MDRO Laboratory Communication

Author: Marious Akugri

Version: 0.4 reviewer-ready draft

Date: 2026-08-25

Purpose

This taxonomy defines common result-status language used across the public artifact suite. The goal is to make laboratory communication more consistent when rapid MDRO-related information is preliminary, pending confirmation, final, corrected, or discrepant.

This document is a reference vocabulary. It does not override local reporting rules, laboratory director approval, medical director approval, public-health reporting requirements, or institutional policy.

Taxonomy Table

Result Status	Working Meaning	Laboratory Communication Risk	Suggested Communication Posture	Scope Note
Preliminary rapid molecular result	A rapid molecular assay has detected an organism, marker, or target before full culture or susceptibility context is complete.	Treating the rapid result as final.	State that the result is from rapid molecular testing and note pending culture, AST, or final report status where applicable.	Do not select therapy.
Organism identification pending confirmation	Initial organism identification may require additional confirmation, referral, or technical review.	Overstating uncertain organism identity.	Use cautious wording and follow local confirmation and reporting pathways.	Especially important for unusual yeast identification and suspected <i>C. auris</i> .
Resistance marker detected	A resistance gene or marker is detected by rapid method.	Equating marker detection with complete phenotypic susceptibility interpretation.	Identify the detected marker and note pending or final AST status.	Local validation and reporting rules control final wording.
AST pending	Susceptibility testing has not been finalized.	Closing the communication loop before final susceptibility information is available.	Communicate rapid or preliminary information with clear AST pending status.	Create follow-up trigger where local process requires it.
AST final	Susceptibility testing is finalized and available for reporting.	Failure to reconcile earlier preliminary comments with final AST.	Review whether earlier comments, alerts, or notifications require update.	Still does not imply treatment direction by the laboratory artifact.
Corrected result	A corrected report changes earlier result information.	Earlier communication may remain in circulation without follow-up.	Document correction, route follow-up under local policy, and update comments if appropriate.	Correction handling must follow institutional procedures.
Discrepant rapid/culture result	Rapid result and culture, AST, or organism identification do not fully align.	Conflicting communication or unsupported certainty.	Escalate for technical review and avoid definitive wording until resolved under local process.	Preserve local quality and technical review requirements.

Result Status	Working Meaning	Laboratory Communication Risk	Suggested Communication Posture	Scope Note
Suspected C. auris	Findings suggest possible C. auris but confirmation is pending or not yet complete.	Premature confirmed-status language or missed infection-prevention awareness route.	Use suspicion/confirmation-pending wording and route awareness according to local policy.	Does not establish a diagnosis or containment order.
Confirmed C. auris	Local or reference method confirms C. auris under approved reporting criteria.	Incomplete communication of status during care transitions.	Follow local infection prevention, public-health, and transfer-communication procedures.	Communication wording must match local approved language.
Negative rapid test with culture pending	Rapid testing does not detect the target, but culture or other final workup remains pending.	Treating a negative rapid test as a final exclusion.	State rapid-test limitation and continue final-result follow-up where applicable.	Avoid closing follow-up prematurely.
Notification documented	A required or selected communication pathway has been completed and recorded.	Missing audit trail.	Record route, time, role/recipient category, and follow-up flag under local policy.	Do not create duplicate documentation outside approved systems.
Follow-up pending	A final, corrected, confirmatory, or AST-related step remains unresolved.	Loss of continuity across shifts or sections.	Maintain a local follow-up queue or flag if approved.	The format depends on the local LIS and quality system.

Status-Aware Comment Pattern

Neutral comments should answer four questions without directing clinical care:

1. What was detected or changed?
2. What method or status applies?
3. What remains pending, if anything?
4. What local follow-up route applies, if any?

Example pattern:

[Method/status] detected [organism, marker, or result change]. [Culture, AST, confirmation, or final report] is [pending/final/corrected] as applicable. Follow institutional notification and follow-up procedures.

Reviewer Questions

- Are any result-status terms missing?
- Are any terms too broad for laboratory communication use?
- Does the taxonomy help distinguish preliminary, final, corrected, discrepant, and confirmation-dependent results?
- Are the C. auris status distinctions cautious enough?
- Would this vocabulary help reviewers evaluate the decision tree and LIS comment examples?

Synthetic Validation Scenarios

Title: Dry-Run Scenarios For Rapid MDRO Diagnostic Stewardship Workflow Review

Author: Marious Akugri

Version: 0.4 reviewer-ready draft

Date: 2026-08-25

Purpose

These synthetic scenarios let reviewers test whether the artifact suite handles realistic laboratory communication edge cases. They are designed for dry-run review only.

The scenarios do not use patient data, employer data, accession numbers, facility identifiers, screenshots, or local SOP language. They do not demonstrate clinical validation or institutional adoption.

In this document, "validation scenario" means a synthetic reviewer exercise. It does not mean validation of a diagnostic assay or validation of a live institutional workflow.

Dry-Run Method

- For each scenario, a reviewer may ask:
1. Which result-status category applies?
 2. Which decision-tree path is triggered?
 3. Which reflex or confirmatory review question applies?
 4. Which LIS comment style would be safe and neutral?
 5. What notification or follow-up documentation would be expected under local policy?
 6. Does the scenario remain within laboratory communication scope?

Scenario Table

Scenario	Synthetic Trigger	Expected Workflow Emphasis	Example Safe Output
S1: Rapid molecular organism and resistance marker	A blood culture rapid molecular panel detects Organism X and Resistance Marker Y. Culture and AST are pending.	Preliminary rapid result, resistance-marker status, final culture/AST follow-up.	Comment notes rapid detection, marker detection, and pending culture/AST; notification follows local policy.
S2: Final AST updates earlier rapid result	Final AST later confirms a resistant phenotype after a preliminary rapid result was communicated.	Final-result reconciliation and follow-up documentation.	Workflow prompts review of whether earlier comments or notifications require update.
S3: Suspected C. auris, confirmation pending	Yeast identification suggests possible C. auris but confirmatory testing or reference review is pending.	Confirmation-aware wording and infection-prevention awareness route if local policy requires it.	Comment avoids confirmed-status language and states that confirmation is pending.
S4: Confirmed C. auris	Approved local or reference method confirms C. auris.	Confirmed-status communication, local public-health/infection-prevention routing, and transfer-status awareness.	Workflow routes communication under local approved procedures without creating treatment or isolation orders.
S5: Corrected organism identification	A final organism identification is corrected after an earlier preliminary communication.	Corrected-result handling, revised communication, and audit trail.	Documentation records corrected result status and follow-up route under local policy.
S6: Negative rapid target with culture pending	Rapid testing does not detect target organisms, but culture remains pending.	Avoiding premature closure.	Comment or workflow notes that rapid result is not the final culture result where local policy requires.

Scenario	Synthetic Trigger	Expected Workflow Emphasis	Example Safe Output
S7: Discrepant rapid and culture result	Rapid result detects a marker, but later culture/AST context does not clearly align.	Technical review before definitive wording.	Workflow prompts supervisory or technical review and avoids unsupported certainty.
S8: Transfer communication reminder	A patient-associated C. auris status may need communication during transfer under local policy.	Status-aware communication without directing transfer decisions.	Comment reminds staff to follow institutional transfer-communication procedures for confirmed or relevant status.

Pass Criteria For Reviewer Assessment

A scenario passes dry-run review if the suite:

- identifies the correct result-status category;
- avoids overstating preliminary or confirmation-pending results;
- routes communication to local policy rather than inventing a universal recipient list;
- preserves laboratory scope and avoids treatment direction;
- prompts final, corrected, AST, or confirmation follow-up where needed;
- produces neutral LIS comment language; and
- leaves local implementation decisions to laboratory leadership and institutional governance.

Failure Modes To Watch

Reviewers should flag any wording that:

- implies clinical treatment selection;
- treats a preliminary rapid result as final;
- labels suspected C. auris as confirmed before confirmation;
- creates mandatory public-health reporting language outside local requirements;
- claims institutional adoption or clinical validation;
- uses patient-specific or employer-confidential details; or
- lacks a final-result or corrected-result reconciliation step.

Reviewer Notes

The goal of these scenarios is practical coherence. A reviewer does not need to agree with every sample comment to find the suite useful; the key question is whether the package provides a safe, adaptable framework for laboratory review and local revision.

Reviewer Feedback Checklist

Title: Structured Technical Feedback For Rapid MDRO Diagnostic Stewardship Artifact Suite

Author: Marios Akugri

Version: 0.4 reviewer-ready draft

Date: 2026-08-25

Purpose

This checklist gives technical reviewers a focused way to evaluate the public artifact suite. It may be used for comments, a review memo, or as preparation for a signed technical review letter if the reviewer is comfortable providing one.

Reviewers should evaluate technical usefulness and safety boundaries only. Reviewers should not make immigration, legal, or adjudicative conclusions.

Reviewers may answer only the sections matching their expertise. A microbiology reviewer, public-health reviewer, and quality/regulatory reviewer do not need to comment on the same issues.

Reviewer Information

Reviewer name:

Professional title:

Organization:

Relevant expertise:

Date of review:

Materials Reviewed

Please mark the materials reviewed.

- ☐ Implementation map and release notes
- ☐ Public technical note
- ☐ Rapid MDRO result-to-action decision tree
- ☐ Reflex and confirmatory testing matrix
- ☐ LIS alert/comment template bank
- ☐ TAT and escalation metric plan
- ☐ Staff training checklist
- ☐ Result-status taxonomy
- ☐ Synthetic validation scenarios
- ☐ Reviewer packet

Evaluation Checklist

Review Area	Rating Or Notes
The suite is understandable to laboratory professionals.	
The proposed workflow is realistic for high-complexity laboratory review.	
The package stays within laboratory communication and diagnostic stewardship support.	
The package avoids treatment guidance and patient-specific clinical direction.	
Preliminary, final, corrected, pending, and confirmation-dependent results are distinguished clearly.	
The C. auris examples are cautious and confirmation-aware.	

Review Area	Rating Or Notes
The LIS comment examples are neutral and adaptable under local governance.	
The reflex/confirmatory testing matrix supports practical review without mandating local testing.	
The TAT and escalation metrics are reasonable for quality review.	
The training checklist and synthetic scenarios are useful for staff education or dry-run review.	
The package avoids patient data and employer-confidential information.	
The package could be adapted by more than one high-complexity clinical laboratory under local approval.	

Short-Answer Questions

1. Which artifact is strongest, and why?
1. Which artifact most needs revision before public release?
1. Are any claims too broad, unsupported, or outside laboratory scope?
1. Are any safety boundaries missing?
1. Would this package be useful as a starting reference for laboratory workflow review, staff training, LIS-comment drafting, or quality discussion?
1. What additional artifact, example, or clarification would make the package more useful?
1. If you are willing to comment on adaptability, could the package be adapted outside one facility under local governance?

Optional Reviewer Confirmation

Optional confirmation language:

I reviewed the public artifact suite listed above. My comments are limited to technical and practical review. I understand that the artifacts are public reference materials, use synthetic examples, and are not presented as approved institutional protocols, treatment guidance, or evidence of institutional adoption.

Return Options

Reviewer feedback may be returned as:

- annotated comments;
- a completed version of this checklist;
- a short review memo; or
- a signed technical review letter, if the reviewer is comfortable providing one.

Reviewer Packet

Title: Technical Review Guide For Rapid MDRO Diagnostic Stewardship Public Artifact Suite

Author: Marious Akugri

Version: 0.4 reviewer-ready draft

Date: 2026-08-25

Purpose

This reviewer packet is for technical reviewers evaluating the public artifact suite titled:

Rapid MDRO Diagnostic Stewardship Workflow and LIS Alert/Comment Reference for High-Complexity Clinical Laboratories

The review should focus on technical coherence, laboratory scope, practical usability, safety boundaries, and adaptability. Reviewers are not being asked to make legal conclusions.

This v0.4 packet is ready for named technical circulation. Reviewers may focus on the lane most relevant to their expertise:

- microbiology, molecular diagnostics, AMR, AST, TAT, and workflow optimization;
- public-health and diagnostic communication;
- laboratory quality, regulatory documentation, specimen/result workflow, and non-confidential boundaries.

Materials For Review

Please review:

1. 00_implementation_map_and_release_notes.md
2. 01_public_technical_note.md
3. 02_rapid_mdرو_result_to_action_decision_tree.md
4. 03_reflex_confirmatory_testing_matrix.md
5. 04_lis_alert_comment_templates.md
6. 05_tat_escalation_metric_plan.md
7. 06_staff_training_checklist.md
8. 07_result_status_taxonomy.md
9. 08_synthetic_validation_scenarios.md
10. 09_reviewer_feedback_checklist.md

Fast Review Path

If time is limited, reviewers can still provide useful technical feedback by following this sequence:

1. Confirm the scope boundary in the implementation map and public technical note.
2. Review the result-status taxonomy.
3. Walk through at least three synthetic scenarios: one rapid molecular/resistance-marker case, one *C. auris* case, and one corrected or discrepant result case.
4. Check whether the decision tree, matrix, and LIS comment examples handle those scenarios without overstating the result or directing treatment.
5. Return either the feedback checklist, a short memo, or a signed review letter limited to the reviewer's own expertise.

Review Scope

Please evaluate whether the artifact suite:

- is technically coherent from a clinical laboratory perspective;
- is appropriately limited to laboratory communication and diagnostic stewardship support;
- avoids prescribing or patient-specific treatment direction;
- distinguishes preliminary, final, corrected, pending, and confirmation-dependent results;
- uses result-status terminology consistently across artifacts;
- uses non-confidential and synthetic examples;
- provides practical communication and documentation logic;
- can be dry-run against synthetic scenarios before any local implementation;
- could be adapted by more than one high-complexity clinical laboratory under local governance; and
- identifies reasonable safety boundaries.

Review Questions

Technical Coherence

1. Does the decision tree reflect a realistic laboratory workflow for rapid MDRO-related findings?
2. Does the matrix handle preliminary, final, corrected, pending, and discrepant results clearly?
3. Are the *C. auris* examples appropriately cautious about confirmation status?
4. Are culture, organism identification, AST, and rapid molecular findings distinguished clearly?
5. Do the synthetic scenarios expose any missing workflow steps?

Laboratory Scope

1. Does the suite stay within laboratory communication and diagnostic stewardship support?
2. Does any section appear to direct clinical care, medication choice, isolation orders, or transfer decisions?
3. Are local policy, laboratory director, medical director, infection prevention, stewardship, quality, and LIS governance roles respected?

Practical Use

1. Are the LIS comment examples clear and neutral?
2. Are the comments adaptable to a local LIS environment after approval?
3. Are the metric categories reasonable for evaluating workflow consistency?
4. Are the staff training cases realistic enough for educational use?
5. Does the reviewer feedback checklist capture the most important practical review questions?

Safety And Confidentiality

1. Does the suite avoid patient data and employer-confidential procedures?
2. Are synthetic examples clearly marked?
3. Are there any claims of adoption, implementation, or institutional approval that should be removed?
4. Are there any sections that need stronger warnings or boundaries?

Adaptability

1. Could a high-complexity clinical laboratory adapt parts of this package under local governance?
2. Which artifact is strongest?
3. Which artifact needs the most revision?
4. What additional artifact would make the package more useful?

Optional Feedback Table

Artifact	Strengths	Concerns	Suggested Revisions
Public technical note			
Decision tree			

Artifact	Strengths	Concerns	Suggested Revisions
Reflex/confirmatory matrix			
LIS template bank			
Metric plan			
Staff training checklist			
Result-status taxonomy			
Synthetic validation scenarios			
Reviewer feedback checklist			

Suggested Reviewer Statement Topics

If a reviewer later chooses to write a signed technical review letter, useful topics may include:

- professional background and relevance to clinical laboratory, microbiology, molecular diagnostics, infection prevention, stewardship, quality, or healthcare operations;
- materials reviewed;
- whether the suite is technically coherent;
- whether the suite is laboratory-scoped and non-prescriptive;
- whether the suite addresses a practical communication gap around rapid MDRO-related results;
- whether the artifacts are public, non-confidential, and adaptable;
- whether the synthetic scenarios and result-status taxonomy make the package easier to review or adapt;
- whether Marious's documented laboratory background appears relevant to the work;
- limitations or revisions recommended; and
- whether continued development and review of the suite would be useful.

The reviewer should avoid legal conclusions, immigration conclusions, unsupported adoption claims, and statements that the package is an approved institutional protocol.

The reviewer should also avoid saying the package has been clinically validated, implemented, or adopted unless they have direct evidence of that fact. A technical review letter can be strong without claiming adoption.

Reviewer Confirmation

Optional confirmation language:

I reviewed the public artifact suite listed above. My comments are limited to technical and practical review. I understand the artifacts are public reference materials, use synthetic examples, and are not presented as approved institutional protocols or treatment guidance.

Return Format

Reviewer feedback may be returned as:

- annotated comments;
- a short review memo;
- completed feedback table; or
- a signed technical review letter, if the reviewer is comfortable providing one.